

**Real-World Order of Operations**Question Bank - Additional 4.07 **ANSWER KEY**

Name

Date

Period

*Show your work. Circle your final answer.***For questions 1 through 4, substitute and evaluate.**

1.  $5x^2$  when  $x = -4$

**80**

$5(-4)^2 = 5(16)$

2.  $\frac{1}{2}mv^2$  when  $m = 30$  and  $v = 10$

**1,500**

$15(100)$

3.  $\frac{a^2+b}{c}$  when  $a = 8$ ,  $b = 4$ ,  $c = 17$

**4***The bar groups the whole numerator:  $64 + 4 = 68$ .*

4.  $3(x+y)^2$  when  $x = -5$  and  $y = 2$

**27**

$3(-3)^2 = 3(9)$

**For questions 5 and 6, write and evaluate an expression. Give the unit.**

5. A square room of side 13 feet is floored at 2.75 dollars a square foot, with a flat 90 dollars for delivery. Find the total cost.

$13^2(2.75) + 90 = 554.75$  **dollars**

6. A rectangular patio 16 feet by 7 feet is sealed at 1.40 dollars a square foot, with a flat 35 dollars for equipment hire. Find the total cost.

$16(7)(1.40) + 35 = 191.80$  **dollars**

*The area is 112 square feet.*

### Two formulas.

7. Body mass index is  $B = \frac{703w}{h^2}$ , where  $w$  is a weight in pounds and  $h$  is a height in inches. Find  $B$  to the nearest tenth.

a.  $w = 120$  ,  $h = 62$

**21.9**

$$\frac{84,360}{3,844} = 21.945\dots$$

b.  $w = 165$  ,  $h = 68$

**25.1**

$$\frac{115,995}{4,624} = 25.085\dots$$

c.  $w = 210$  ,  $h = 74$

**27.0**

$$\frac{147,630}{5,476} = 26.959\dots$$

8. Kinetic energy is  $K = \frac{1}{2}mv^2$ , for mass  $m$  in kilograms and speed  $v$  in metres per second. Find  $K$  in joules.

a.  $m = 1,500$  ,  $v = 20$

**300,000 joules**

b.  $m = 640$  ,  $v = 9$

**25,920 joules**

c.  $m = 55$  ,  $v = 4$

**440 joules**

### Working backwards.

9. Find the missing value in each situation.

a.  $\frac{1}{2}mv^2 = 1,600$  and  $v = 8$  . Find  $m$  .

**$m = 50$**

$$32m = 1,600$$

b.  $x^2 - 7 = 74$  , where  $x$  is positive. Find  $x$  .

**$x = 9$**

- c. A square lawn of side  $n$  feet is turfed at 3.00 dollars a square foot with a flat 55 dollars delivery, and the total is 1,255 dollars. Find  $n$  .

**$n = 20$**

$$n^2(3.00) = 1,200 \text{ , so } n^2 = 400$$

**Reading the situation.**

10. Two problems use the numbers 8, 12 and 4. **A:** a printer costs 8 dollars to hire plus 12 dollars for each of 4 days. **B:** a printer costs 8 dollars for each of 12 reams, plus 4 dollars for delivery. Find both totals, and explain why circling the numbers would not have told you what to do.

**A is  $8 + 12(4) = 56$  dollars. B is  $8(12) + 4 = 100$  dollars. The two problems carry the same three numbers, so the numbers cannot be what decides; what decides is which amount is repeated and how often.**

*Ask which number is a rate in each problem — that question has a different answer in A and in B, and no marking on the page can show it.*

11. A square court of side 20 feet is resurfaced at 3.60 dollars a square foot, with a flat 200 dollars for the equipment. Which expression gives the total cost?

- ☒ **A.**  $20^2(3.60) + 200$   
☐ B.  $20^2(3.60 + 200)$   
☐ C.  $20(3.60) + 200$   
☐ D.  $20^2 + 3.60 + 200$

**A**

12. Select ALL of the expressions whose value is 32 when  $x = 4$ .

- ☒ **A.**  $2x^2$   
☐ B.  $(2x)^2$   
☒ **C.**  $x^2 + 16$   
☒ **D.**  $\frac{128}{x}$   
☒ **E.**  $x^2 \cdot 2$   
☐ F.  $2x \cdot 2$

**A, C, D and E**